Autonomous Pipe-Inspection Rover — Bill of Materials

Yunus Hassen · George Mason University · Rev. 2 — 28 September 2026 · Prototype (v1) build · Project budget cap $500.00

1 · TO PURCHASE — PROTOTYPE BUILD

Item	Specification / Purpose	Qty	Ext. Cost
Raspberry Pi Zero 2 W	Camera compute node — captures distance-triggered stills and forward video. **Long-lead item: stock-constrained.**	1	$15.00
Arducam B0310	12 MP IMX708, 120° M12 manual-focus module — primary inspection camera. Fixed focus set to pipe-wall distance.	1	$36.99
microSD — A2 64 GB	Pi Zero 2 W boot + image storage. A2 random-write rating.	1	$12.00
microSD — A1 32 GB	ESP32 telemetry log (distance, pitch, roll, timestamps).	1	$6.00
microSD SPI breakout	Card interface for the ESP32. Sold in 3-packs.	3-pk	$8.00
TMC2209 driver	Gimbal pan stepper driver. StealthChop for low noise; set Vref before first run. 5-pack covers drive motors later.	5-pk	$20.00
Pololu D24V22F5	5 V / 2.5 A switcher, 85–95% efficient. Power-good pin drives brownout warning to ESP32.	1	$15.00
LM2596 buck module	Adjustable regulators for the servo and auxiliary rails. Multipack of 8–10.	multi	$10.00
1000 uF electrolytic	Bulk capacitance on the servo rail. 25 V rating. Sold in assortment kits.	kit	$10.00
Subtotal — Section 1			**$132.99**

2 · LIDAR SCANNING ASSEMBLY + CHASSIS — CONFIRM PURCHASE OWNER

Item	Specification / Purpose	Qty	Ext. Cost
Benewake TFmini-S	Forward ToF lidar, 12 m, UART 115200. Driver written and simulated. DFRobot $39.90; getFPV $44.99.	1	$39.90
NEMA 14 stepper	**Gimbal pan axis** (360° front-to-back). Angle-tags every lidar sample. StepperOnline 14HM11-0404S, 0.9°, 11 Ncm, 0.4 A.	1	$13.00
MG90S servo	**Gimbal tilt axis** (±45°). Metal gear required — SG90 will not survive a 60–90 min run. 2-pack incl. spare.	2-pk	$11.00
PLA filament	Chassis, camera mounts, gimbal housing. Overture ~$15/kg; Hatchbox $21.99/kg.	3 kg	$45.00
Subtotal — Section 2			**$108.90**

3 · ON HAND — NO COST

Item	Specification / Purpose	Qty	Ext. Cost
ESP32 DevKitC	Real-time controller: stepper, servo, lidar, IMU, telemetry logging.	1	$0.00
ESP32-S3-Zero	Secondary controller, reserved for motor/encoder expansion.	1	$0.00
MPU-6050 IMU	6-axis. Roll feedback for pipe-invert centering; pitch for grade profiling.	1	$0.00
OV5647 camera + FFC	SunFounder module, 15-to-22 pin cable. Forward-view video camera.	1	$0.00
Capacitors 0.1–220 uF	Covers TMC2209 bulk-cap requirement and per-IC decoupling.	assort	$0.00

4 · NOT YET BUDGETED — DEFERRED, ESTIMATES ONLY

Item	Specification / Purpose	Qty	Ext. Cost
Drive motors — GAP	**No propulsion specified.** Both Section 2 motors are gimbal axes. Needs geared DC motors w/ encoders or a stepper pair.	2	$40–80
Wheel encoders — GAP	Step count no longer gives odometry. Required for distance stamps and photo trigger. **Exceeds ESP32 pin budget** — note 2.	2	$15–30
LED array + CC driver	CRI ≥ 90, 4000–5000 K, constant-current. Quantity set by the cardboard-tube lumen test.	—	$30–60
Battery — 3S Li-ion	11.1 V, 30–40 Wh target. **Order last** — sized after lighting draw is measured.	1	$30–50
Balance charger	Required for any 3S pack. Commonly overlooked line item.	1	$30–40
Drivetrain / wheels	Wheels, couplers, bearings. Profile must match the curved pipe invert.	—	$50–100
Housing, seals, mounts	Enclosure; thermal path from LEDs and compute into the housing wall.	—	$30–80
Wire, connectors, spares	Silicone wire, JST/XT60, 2.2 kΩ I2C pull-ups, replacement stock.	—	$80
Estimated range — Section 4			**$305–520**

Sections 1 + 2	**$241.89**	**Budget cap**	**$500.00**	**Remaining**	**$258.11**

Notes. (1) **Projected total $547–762 against a $500 cap.** Sections 1 and 2 are affordable; Section 4 is not yet funded and must be reduced or the cap raised. (2) With the NEMA 14 assigned to gimbal pan, step count no longer serves as odometry. Wheel encoders become mandatory, and the ESP32 is already at ~21 of ~25 usable GPIO — motor and encoder duties should move to the ESP32-S3-Zero. (3) Confirm who purchases Section 2 and whether it draws on the same $500 — do not double-order. (4) A second Pi Zero 2 W is desirable for independent camera failure domains; buy immediately if found at list price. (5) Prices are single-unit retail as of 28 Sep 2026 and exclude shipping and tax.